Ujjwal Prajapati

Bhubaneswar, Odisha

+91 8957408434 Email LinkedIn GitHub LeetCode Portfolio

Summary

Computer Science undergraduate with hands-on experience in software development, REST APIs, testing, debugging, and real-time systems. Proficient in Java, C++, Python, JavaScript, and TypeScript, with a strong foundation in Data Structures and Algorithms, Object-Oriented Programming, Software Engineering, Databases, and Computer Networks. Experienced in designing modular software components, developing and testing APIs, troubleshooting application issues, optimizing software performance, and collaborating through Git-based development workflows.

Education

XIM University **Aug 2023 – Present**
*B.Tech in Computer Science — **Expected 2027 — CGPA: 9.16/10*** *Bhubaneswar, Odisha*

Kendriya Vidyalaya Faizabad Cantt **2019 – 2021**
Higher Secondary (CBSE) *Uttar Pradesh*

Experience

PixScripti Technologies **May 2026 – June 2026**
Technical Intern – Frontend Development *Remote*

- Collaborated on **RESTful API integration and JSON payload design** to support reliable asynchronous communication between application components.
- Translated backend database schemas into predictable client-side data models and optimized state-management workflows for application features.
- Implemented and debugged authentication flows, route guards, and dynamic data handling while resolving asynchronous race conditions and application issues.
- Performed **API integration and contract testing** to identify data inconsistencies, rendering failures, race conditions, and client-server communication issues.
- Used **Git, GitHub, code reviews, and Agile workflows** to collaborate on application development and maintain reusable components.

Projects

CrowdShield AI – Real-Time Crowd Safety Platform | *Next.js, React Native, TypeScript, FastAPI, WebSockets, Supabase***Git**

- Developed a real-time crowd safety platform integrating telemetry ingestion, risk analysis, alert generation, and operator response workflows across web and mobile applications.
- Designed modular real-time inference and event-delivery components for processing crowd telemetry and delivering risk updates and alerts to operator dashboards through REST APIs and WebSockets.
- Implemented temporal risk analysis and historical intelligence to identify changing crowd conditions and support early-warning workflows.
- Built operator-facing interfaces for real-time crowd state, camera health, risk trends, alerts, and incident timelines.
- Validated backend functionality through **298 automated tests with 0 failures** and maintained TypeScript compilation with **0 errors**.

VTRACE – Algorithm Visualization Tool | *React, Next.js, CSS Animations* **GitHub**

- Designed and developed an interactive software tool for visualizing **algorithms, data structures, and state transitions** in real time.
- Implemented algorithm execution logic and component state management to support pause, play, and step-by-step execution workflows.
- Optimized component rendering and state boundaries to efficiently handle rapid visual updates during large algorithm simulations.

Billboard Detection & Reporting Portal | *React, Next.js, Tailwind CSS, Web APIs* **GitHub**

- Developed a role-based reporting application with browser-based **geolocation capture, image upload, and record management** functionality.
- Designed reusable components and stateful workflows for administrative operations including data tables, search, filtering, and record management.
- Implemented client-side validation and application logic to maintain reliable data flow and reduce unnecessary network requests.
- Debugged state, validation, and rendering issues while optimizing client-side application performance.

Technical Skills

Programming Languages: Java, C++, C, Python, TypeScript, JavaScript
Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Algorithm Design & Analysis, Computer Networks, Database Management Systems
Software Engineering: Software Design, Testing, Debugging, REST APIs, Git, Code Reviews, Agile Development
Backend & Systems: FastAPI, WebSockets, Docker
Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS
Databases: MongoDB, PostgreSQL, SQLite
Tools & Platforms: Git, GitHub, VS Code, Vite, Supabase, Figma

Leadership & Achievements

- **IEEE XIM CS Student Branch – Webmaster & Technical Team:** Supported the student branch's web presence and technical initiatives while collaborating with student teams on technology-driven activities.
- **2nd Runner Up** in Technova Hackathon powered by ESSPL (Competed as a Solo Finalist).
- **Merit Scholarship Recipient** – Academic Excellence.
- **CS50x: Introduction to Computer Science** – Harvard University (2024) Link

Coding Profiles

- **LeetCode:** Solved **900+ problems** across algorithms and data structures; maintained an **800+ day coding streak**.
- **GitHub:** 10+ public repositories containing software projects, backend and frontend applications, real-time systems, interactive tools, and visualizations.